

# Machine-checked rooted-tree majorants for polymer expansions with holes

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## Abstract

We present a machine-checked quantitative toolkit for cluster expansions of polymer systems with excluded regions (*holes*), in the discrete cube geometry of Balaban–Dimock renormalization-group analyses. Five Lean 4 theorems, checked against a pinned Mathlib revision, provide: (i) the identity  $\sum_T \prod_v c_T(v)! = n! C_n$  for child factorials over spanning trees of the complete graph  $K_{n+1}$ , with the rooted-tree majorant  $4^n$  as corollary; (ii) a marked-root leaf summation for the tree-graph majorant of an Ursell-type expansion with holes, with the moment constant  $M$  paid once at the root and closed leaf ratio  $4M^2$  per additional vertex, together with its Catalan-sharpened form  $M^{2n+1}C_n$  (a gain of order  $n^{3/2}$  in the  $n$ th coefficient); and (iii) a target-preserving orderwise bound in which the target union itself survives until the modified-metric exponential is extracted. A certified companion (interval arithmetic, 120-bit precision, committed transcript with a committed reproduction witness) tabulates the smallness gate and encloses every derived constant. Non-vacuity is machine-checked: a concrete hole family satisfying every hypothesis is exhibited in Lean, and the two distinct hypothesis sets among the polymer-facing theorems are both instantiated at it with a strictly positive weight. Each claim is labelled with its verification layer: exact (Lean theorem), certified (interval transcript), or paper-level.

## 1 Introduction

Cluster (polymer) expansions convert log-partition functions into sums over connected geometric objects; their convergence proofs hinge on combinatorial majorants for tree sums and on geometric smallness gates [1, 2, 3]. When the expansion must respect *holes* — excluded regions that the expansion sees as atomic units, as in renormalization-group treatments of large-field regions [6, 8] — the bookkeeping is notoriously error-prone: root normalizations, child factorials, and hole-separation hypotheses interact in ways that invite wrong constants.<sup>1</sup>

This paper contributes, in machine-checked form:

1. an exact Catalan evaluation of the rooted child-factorial tree sum (Theorem 3.1), of which the classical  $4^n$  tree-shape budget (Theorem 3.2) is a corollary;

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\*This manuscript and parts of the formal development were produced with AI assistance under the repository’s audit protocol. Every theorem below is machine-checked in Lean 4 against a pinned Mathlib; the constant tables of Section 5 are enclosed by committed interval arithmetic; remaining numerics are labelled paper-level where they occur. Repository: <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>.

<sup>1</sup>A wrong constant of exactly this kind occurred in our own pre-registered prediction of the smallness-gate crossover; the certified run caught it (Remark 5.1).

2. finite, volume-free bounds for the marked-root and fixed-target-union tree-graph majorants of an Ursell-type expansion with holes (Theorems 3.3, 3.4, 3.5), with one smallness gate as the only inequality hypothesis and every constant an explicit function of  $(d, \kappa_0)$ ;
3. a certified constant table (Section 5) and a machine-checked non-vacuity witness (Section 7).

Statuses are kept separate throughout: **exact** means a Lean theorem in the pinned core with axiom oracle `[propext, Classical.choice, Quot.sound]`; **certified** means a committed interval-arithmetic transcript with ball-plus-boolean output; **paper-level** means ordinary mathematical prose not formalized here, and is always flagged. The formal development lives in a larger repository whose top-level namespace (`YangMills/`) reflects its original motivation in lattice gauge theory; the present results are self-contained combinatorial and expansion-theoretic tooling, and no reduction to any continuum or gauge-theoretic problem is claimed or used.

## 2 Setting and definitions

Throughout,  $d, L \in \mathbb{N}$  with  $L \neq 0$ .

**Definition 2.1** (cubes and adjacency). *A cube is a point of the discrete torus  $(\mathbb{Z}/L)^d$  (Lean: `Cube d L`). Two cubes are adjacent (`cubeAdj`) iff they are distinct and each coordinate differs by 0 or  $\pm 1$ ; this is the sup-metric neighbor graph, in which every cube has  $3^d - 1$  neighbors. A cube set is connected when it is walk-connected in this graph.*

**Definition 2.2** (hole families and skeletons). *A hole family  $\mathcal{H}$  (Lean: `HoleFamily d L`) is a finite set of finite cube sets — the connected components of the excluded region. A cube set  $X$  respects  $\mathcal{H}$  iff every hole  $H \in \mathcal{H}$  satisfies  $H \subseteq X$  or  $H \cap X = \emptyset$ : the expansion sees each hole as an atomic unit [8, eq. (149)]. The skeleton  $\text{sk}(X)$  is the set of cubes of  $X$  lying in no hole. The bounds below hypothesize: pairwise disjointness of holes, no adjacency edge between distinct holes, and non-emptiness of each hole.*

**Definition 2.3** (polymers). *Given  $\mathcal{H}$  and an activity  $z: \mathcal{P}((\mathbb{Z}/L)^d)_{\text{fin}} \rightarrow \mathbb{C}$ , a polymer is a nonempty, connected, hole-respecting cube set with nonempty skeleton;  $\Omega$  denotes the (finite) type of polymers (Lean: `OmegaPolymerType`). The activity  $z$  parameterizes the polymer type for compatibility with the upstream expansion; no result below depends on it, and the non-vacuity witness of Section 7 takes  $z \equiv 0$ . Two polymers are incompatible, written  $X \sim X'$ , iff their skeletons intersect. For a tuple  $\mathbf{X} = (X_0, \dots, X_n) \in \Omega^{n+1}$ , its incompatibility graph  $G(\mathbf{X})$  has vertex set  $\{0, \dots, n\}$  and an edge  $\{i, j\}$  iff  $X_i \sim X_j$ .*

**Definition 2.4** (modified metric and hole weight). *The discrete modified metric  $\rho(X)$  (Lean: `discreteModifiedMetric`) is the Steiner length of the skeleton inside  $X$ :*

$$\rho(X) = \min\{|S| - 1 : \text{sk}(X) \subseteq S \subseteq X, S \text{ connected}\},$$

*with  $\rho(X) = 0$  if no such  $S$  exists. For a rate  $\kappa \geq 0$  the hole weight is  $W_\kappa(X) = e^{-\kappa(\rho(X)+1)}$  (Lean: `appendixFHoleExpWeight`).*

**Definition 2.5** (the two tree-graph majorants). Fix  $\mathcal{H}$ , an activity  $z$ , a nonnegative weight  $w: \Omega \rightarrow \mathbb{R}_{\geq 0}$ , and  $n \in \mathbb{N}$ . Write  $\tau(\mathbf{X})$  for the number of spanning trees of  $G(\mathbf{X})$ . The marked-root sum at a cube  $r$  is

$$S_{w,r}^\#(n) = \frac{1}{(n+1)!} \sum_{\substack{\mathbf{X} \in \Omega^{n+1} \\ r \in \text{sk}(X_0)}} \tau(\mathbf{X}) \prod_{j=0}^n w(X_j)$$

(Lean: `appendixFHoleHsharpWeightedTreeMarkedRootSum`), and the fixed-target sum at a finite cube set  $Y$  is

$$T_{w,Y}^\#(n) = \frac{1}{(n+1)!} \sum_{\substack{\mathbf{X} \in \Omega^{n+1} \\ \bigcup_j X_j = Y}} \tau(\mathbf{X}) \prod_{j=0}^n w(X_j)$$

(Lean: `appendixFHoleHsharpWeightedTreeTerm`). These are the standard tree-graph majorants of the order- $(n+1)$  coefficients of an Ursell (Mayer) expansion: by tree-graph inequalities of Penrose type [4, 5], Ursell coefficients are dominated by spanning-tree counts of incompatibility graphs, and it is the majorant objects  $S^\#, T^\#$  — not the signed coefficients — that the theorems below bound. The passage from signed coefficients to these majorants lives upstream in the repository’s Kotecký–Preiss/Mayer–Ursell development and is not re-proved here. (The word “second” in the Lean names refers to the second cluster-expansion pass of the upstream development; nothing in this paper depends on the first.)

**Definition 2.6** (gate and derived constants). The smallness gate at dimension  $d$  and rate  $\kappa_0 > 0$  is

$$G(d, \kappa_0) = (3^d)^2 e^{-\kappa_0} 2^{3^d+1} < 1 \quad (\text{Lean hypothesis } h_{C_q}), \quad (1)$$

and the derived constants are

$$\text{RS} = (1 - G)^{-1}, \quad M = \max(1, \kappa_0^{-1}, \text{RS}), \quad \Lambda = 4M^2 \quad (2)$$

(Lean: `appendixFSecondUrsellMomentConstant`, `appendixFSecondUrsellLeafConstant`). All three are explicit functions of  $(d, \kappa_0)$  alone — in particular volume-free: no constant below depends on the torus size  $L$ , the target  $Y$ , or on  $n$  beyond the displayed powers. In applications where the order- $n$  term additionally carries an activity factor  $\varepsilon^n$ , the resulting geometric series has ratio  $\Lambda\varepsilon$ , which is below one exactly for  $\varepsilon < \varepsilon^* := \Lambda^{-1}$  (strict; nothing is claimed at equality). Status: the Lean endpoints prove the coefficient bounds  $M\Lambda^n$  and  $M^{2n+1}C_n$  (exact); the enclosures of  $\Lambda^{-1}$  are certified; the passage from coefficient bound to a convergent activity series is paper-level algebra conditional on the stated  $\varepsilon^n$  structure.

### 3 Main results

All five statements carry stable, publication-facing names in module `MarkedRootedClosure`, file `PaperTheorems.lean`; each is a direct application of a pinned upstream theorem — no new axiom, no opaque assumption. Here  $\mathcal{T}(K_{n+1})$  is the set of spanning trees of the complete graph on the vertices  $\{0, \dots, n\}$ , each tree rooted at 0, and  $c_T(v)$  is the number of children of  $v$  in the rooted tree  $T$ .

**Theorem 3.1** (exact Catalan value; `normalizedRootedChildFactorialTreeCatalanIdentity`). For every  $n \in \mathbb{N}$ ,

$$\frac{n+1}{(n+1)!} \sum_{T \in \mathcal{T}(K_{n+1})} \prod_v (c_T(v))! = C_n, \quad \text{equivalently} \quad \sum_T \prod_v c_T(v)! = n! C_n,$$

where  $C_n = \binom{2n}{n}/(n+1)$  is the  $n$ th Catalan number.

**Theorem 3.2** (rooted-tree majorant; `normalizedRootedChildFactorialTreeBound`). For every  $n$ , the left-hand side of Theorem 3.1 is at most  $4^n$ .

Theorem 3.2 follows from Theorem 3.1 via  $C_n \leq 4^n$ ; the classical budget overcounts by exactly the factor  $4^n/C_n$ , which is  $\sim \sqrt{\pi} n^{3/2}$ .

**Theorem 3.3** (marked-root leaf summation; `markedRootLeafGeometricBound`). Let  $\mathcal{H}$  satisfy the separation hypotheses of Definition 2.2, let  $w \geq 0$  be dominated by the hole weight at doubled rate,  $w \leq W_{2\kappa_0}$  pointwise, let  $r$  be any cube, and assume the gate (1). Then for every  $n$ ,

$$(n+1) \cdot S_{w,r}^\#(n) \leq M \cdot \Lambda^n.$$

**Theorem 3.4** (Catalan-sharpened marked-root bound; `markedRootCatalanBound`). Under exactly the hypotheses of Theorem 3.3,

$$(n+1) \cdot S_{w,r}^\#(n) \leq M^{2n+1} \cdot C_n.$$

Since  $M\Lambda^n = M^{2n+1}4^n$ , this improves Theorem 3.3 by the factor  $4^n/C_n$ : the tree-shape factor is paid at its exact Catalan value.

**Theorem 3.5** (target-preserving orderwise bound; `targetPreservingWeightedTreeBound`). Let  $\mathcal{H}$  satisfy the separation hypotheses of Definition 2.2 and assume the gate (1). Let  $w, u \geq 0$  with  $u \leq W_{2\kappa_0}$  pointwise, suppose  $w \leq W_{\text{rate}} \cdot u$  pointwise for some rate  $\geq 0$ , and let  $Y$  be a finite cube set whose skeleton contains some cube  $r$ . Then

$$T_{w,Y}^\#(n) \leq M \cdot W_{\text{rate}}(Y) \cdot \Lambda^n.$$

Here “orderwise” means: for each fixed order  $n$  separately.

## 4 Proof architecture

The formal proofs are the authority; this section gives the mathematical mechanism at a level where a specialist can reconstruct the logic. Module pointers are to the repository.

### 4.1 The Catalan identity

The sum  $\sum_T \prod_v c_T(v)!$  counts pairs  $(T, \sigma)$  where  $T$  is a spanning tree of  $K_{n+1}$  rooted at 0 and  $\sigma$  assigns to each vertex a linear order on its children — i.e., it counts *labelled rooted plane trees* on the vertex set  $\{0, \dots, n\}$  with root 0. Plane trees are rigid (a plane structure admits no nontrivial automorphism fixing the root), so this count factors as

$$\#\{\text{plane shapes with } n+1 \text{ nodes}\} \times \#\{\text{labellings of the } n \text{ non-root nodes}\} = C_n \cdot n!,$$

the first factor by the classical bijection between plane trees and binary trees, formalized in `YangMills/KP/PlaneTreeCatalan.lean` through Mathlib’s

`Tree.treesOfNumNodesEq_card_eq_catalan;`

the identity then normalizes to Theorem 3.1 (`YangMills/KP/RootedCatalanExact.lean`).

**Example 4.1.**  $n = 2$ :  $K_3$  has three spanning trees. Rooted at 0, the two paths with 0 as an endpoint contribute  $\prod_v c_T(v)! = 1$  each, and the path through 0 — the star at 0 when rooted — contributes  $2! = 2$ ; the sum is  $4 = 2! C_2$ .

## 4.2 Leaf summation and the $2n + 1$ moment budget

Theorems 3.3 and 3.4 are proved by splitting the tuple sum by the (rooted) shape of a spanning tree and peeling leaves. Three mechanisms produce the constants:

*One-pin polymer sums.* The basic estimate controls the sum of  $W_{2\kappa_0}(X)$  over polymers  $X$  whose skeleton contains a fixed cube. A polymer with  $\rho(X) + 1 = m$  has a skeleton of at most  $m$  cubes carried by a minimal connected Steiner set  $S$ ; enumerating rooted connected Steiner sets cube by cube costs  $(3^d)^2$  per step (degree-squared branching in the adjacency graph), and each cube of  $S$  carries a fiber entropy  $2^{3^d+1}$  (the factor  $2^{|S|}$  for skeleton-subset choices times  $2^{3^d|S|}$  for admissible hole fillings over  $S$ ). The sum over polymers at Steiner length  $m$  is therefore dominated by  $G^m$  with  $G$  the gate (1), and the total by the geometric sum  $RS = (1 - G)^{-1} \leq M$ . The gate is the only inequality *hypothesis*; the constants are then explicit functions of  $(d, \kappa_0)$ , with the  $\kappa_0^{-1}$  component of  $M$  paid by the spare  $\kappa_0$  of decay left over when the doubled-rate weight  $W_{2\kappa_0}$  surrenders only  $W_{\kappa_0}$  to the geometric mechanism.

*Moment paid once at the root.* In the marked-root sum the coordinate  $X_0$  is pinned at  $r$ , so the root contributes one one-pin sum: one factor  $M$ . Each tree edge propagates a pin: a child polymer must have skeleton intersecting its parent's, so summing the child costs one one-pin sum ( $M$ ) *after* choosing the pin inside the parent, whose cost is accounted to the parent's own moment. The bookkeeping closes with each of the  $n$  non-root vertices paying two factors ( $M$  for its own one-pin sum,  $M$  for the pin it offers its children) and the root paying one:  $M^{2n+1}$  total. The formal count is the fixed-parent kernel lemma

`appendixFHoleHsharpWeightedTreeMarkedRootFixedParentKernelSum`  
`_le_prod_childMomentBudget_mul_rootMoment`

and its consumers in `YangMills/RG/AppendixFSecondUrsellLeafSummation.lean`.

*Shape sum.* What remains is the sum over rooted tree shapes with child multiplicities — exactly the object of Section 4.1. Paying it at its Catalan value gives Theorem 3.4; paying it at the  $4^n$  budget and absorbing  $4^n M^{2n} = \Lambda^n$  gives Theorem 3.3. The  $(n + 1)$  prefactor in both statements is the marked-coordinate count: pinning coordinate 0 rather than an arbitrary coordinate costs a factor  $n + 1$  against the  $(n + 1)!$  normalization.

## 4.3 Target preservation

In Theorem 3.5 the tuple union is constrained to a fixed target  $Y$ . Extracting the metric exponential *before* summing would discard this constraint and with it the geometry of  $Y$ . Instead, the hypothesis  $w \leq W_{\text{rate}} \cdot u$  is applied per polymer, and the extracted factors combine, for each contributing tuple (connected in the incompatibility graph; disconnected tuples contribute zero to  $T^\#$ ), into at least the full-target factor  $W_{\text{rate}}(Y)$ : the superadditivity  $\rho(Y) + 1 \leq \sum_i (\rho(X_i) + 1)$  is proved by stitching minimal Steiner sets along a spanning tree of the incompatibility graph (Lean: `omegaClusterUnion_discreteModifiedMetric_add_one_le`

`_sum_of_spanningTree`). Only after this factor is secured is the remainder  $u$  routed through the marked-root bound at a root  $r \in \text{sk}(Y)$ . The order of operations is the content of the theorem’s name and the reason the constant is volume-free in  $Y$ .

## 5 The certified companion

Script `scripts/c1_admissibility_arb.py` (SHA-256

`f114fc4bcf2193c47603fb595b91b4e3ea62addc3a1bfebe0aa0c147c447f1f9`

`python-flint` 0.9.0, precision 120 bits; transcript committed alongside it, exit 0) encloses  $G$ ,  $M$ ,  $\Lambda$  (printed as `Lf` in the transcript), and  $\varepsilon^*$  with midpoint-radius enclosures plus provable booleans; the gate booleans are tri-state and provable-only (PASS only when  $G < 1$  holds for every point of the enclosing ball, FAIL only when  $G \geq 1$  does, UNDECIDED otherwise — no UNDECIDED occurs in the table). The run was re-executed from scratch on 2026-07-11 and reproduced the committed transcript byte-identically; the comparison log (SHA-256 of both transcripts) is committed as `scripts/c1_admissibility_rerun_20260711_COMPARISON.txt`. One header note: the script docstring’s phrase “certified activity radius” predates the status correction of Definition 2.6, which governs. Selected rows (midpoints shown; radii in the transcript, all below  $3 \times 10^{-6}$  relative):

| $d$ | $\kappa_0$ | gate $G$              | $M$                       | $\Lambda$          | $\varepsilon^*$       |
|-----|------------|-----------------------|---------------------------|--------------------|-----------------------|
| 2   | 11         | 1.3853059             | FAIL (provably $\geq 1$ ) |                    |                       |
| 2   | 11.5       | 0.84023048            | 6.2590162                 | 156.70114          | 0.0063815746          |
| 2   | 12         | 0.50962555            | 2.0392580                 | 16.634292          | 0.060116775           |
| 2   | <b>16</b>  | 0.0093341175          | 1.0094221                 | 4.0757316          | <b>0.24535472</b>     |
| 2   | 24         | $3.13 \times 10^{-6}$ | 1.0000031                 | 4.0000251          | 0.24999843            |
| 3   | 25         | 2.7177241             | FAIL (provably $\geq 1$ ) |                    |                       |
| 3   | 26         | 0.99979481            | 4873.5020                 | $9.50 \times 10^7$ | $1.05 \times 10^{-8}$ |
| 3   | 28         | 0.13530751            | 1.1564805                 | 5.3497886          | 0.18692327            |
| 3   | <b>33</b>  | 0.00091169486         | 1.0009125                 | 4.0073035          | <b>0.24954436</b>     |

The certified crossovers of the gate are

$$\kappa^*(2) \in 11.32592096 \pm 2.72 \times 10^{-10}, \quad \kappa^*(3) \in 25.99979479 \pm 2.32 \times 10^{-9},$$

and  $\varepsilon^* \uparrow 1/4$  as  $\kappa_0 \rightarrow \infty$  (the pure tree-majorant limit  $\Lambda \downarrow 4$ ). The row  $(d, \kappa_0) = (3, 26)$  sits just above  $\kappa^*(3)$  and displays the blow-up of the derived constants as  $G \rightarrow 1^-$ : admissible, but with  $M \approx 4.9 \times 10^3$  and  $\varepsilon^* \approx 1.05 \times 10^{-8}$ .

Scope of the certificates: the transcript’s band booleans certify the gate at the two registered band points  $\kappa_0 = 16$  and  $\kappa_0 = 33$ ; the extension to all larger  $\kappa_0$  is the elementary monotonicity of  $\kappa_0 \mapsto e^{-\kappa_0}$  (paper-level, one line). Admissibility everywhere means the strict gate  $G < 1$ , as in the Lean hypothesis.

**Remark 5.1** (a measured registration failure). *Before the certified run, we committed to the repository (`docs/C1-CHARTER.md`) a predicted crossover for the gate: 15.72 ( $d=2$ ) and 32.60 ( $d=3$ ), from a hand-assembled expression carrying  $(3^d)^4$  where the gate (1) carries  $(3^d)^2$ . The certified run measured the discrepancy (transcript, “crossover audit”: REGISTERED PREDICTION*

DOES NOT RE-DERIVE); the pre-committed admissibility bands  $\kappa_0 \geq 16$  ( $d=2$ ) and  $\kappa_0 \geq 33$  ( $d=3$ ) remain true — sufficient, not sharp — and only certified values appear in this paper. We record the failure as evidence for the correction protocol: constants are computed, never assembled by hand.

## 6 What the Catalan sharpening buys

The gain of Theorem 3.4 over Theorem 3.3 at order  $n$  is the integer ratio  $4^n/C_n$  (computed in integer arithmetic, displayed rounded; paper-level, independently recomputable):

| $n$ | $C_n$                         | $4^n$                         | $4^n/C_n$ |
|-----|-------------------------------|-------------------------------|-----------|
| 1   | 1                             | 4                             | 4.00      |
| 2   | 2                             | 16                            | 8.00      |
| 5   | 42                            | 1,024                         | 24.38     |
| 10  | 16,796                        | 1,048,576                     | 62.43     |
| 20  | 6,564,120,420                 | $\approx 1.10 \times 10^{12}$ | 167.50    |
| 30  | $\approx 3.81 \times 10^{15}$ | $\approx 1.15 \times 10^{18}$ | 302.21    |

Being  $\sim \sqrt{\pi} n^{3/2}$ , the improvement is polynomial: it does *not* move the exponential threshold. In an application with per-vertex activity  $\varepsilon$ , both bounds give the same radius requirement  $4M^2\varepsilon < 1$  asymptotically; what changes is every finite-order coefficient, hence every truncation-error estimate at fixed order. Concretely, at  $(d, \kappa_0) = (2, 16)$  (certified row:  $M = 1.0094221$ , ball radius  $\leq 4 \times 10^{-8}$ ), truncating so that order  $n = 10$  is the last term retained, the first omitted coefficient is smaller by the factor  $4^{11}/C_{11} \approx 71.4$  under the Catalan form — almost two decimal digits of truncation error at no analytic cost. For the near-critical certified row  $(3, 26)$ , where  $M \approx 4.9 \times 10^3$  inflates every coefficient through  $M^{2n+1}$ , the same polynomial factor directly divides the order- $n$  term of any error budget. These consequences are paper-level arithmetic on top of the machine-checked theorems.

## 7 Non-vacuity, machine-checked

A characteristic failure mode of formalized bounds — one this programme has itself committed and repaired in earlier work — is hypotheses that no concrete geometry satisfies. Module `MarkedRootedClosure/NonVacuity.lean` therefore exhibits — as instance terms, not existentials — the hole family  $\mathcal{H}^w$  in  $d = 2$ ,  $L = 8$  with two singleton holes at sup-distance 4, and proves by kernel decision (`decide`) that the holes are disjoint, edge-separated, and nonempty; the gate (1) at  $\kappa_0 = 16$  is proved in Lean via  $e^{16} > 2.7^{16} > 82944$  — the Lean-exact counterpart of the certified row  $G \approx 0.00933$ . The two distinct hypothesis sets among Theorems 3.3–3.5 are then both instantiated at  $\mathcal{H}^w$  with the *strictly positive* weight  $W_{32} = e^{-32(\rho+1)}$  (no zero-weight degeneracy; Theorem 3.4 shares its hypotheses verbatim with Theorem 3.3, so its satisfiability follows):

```
nonvacuous_markedRootLeafGeometricBound, nonvacuous_targetPreservingWeightedTreeBound.
```

What this witness proves: both hypothesis sets among Theorems 3.3–3.5 are satisfiable by a concrete geometry with positive weights, so the theorems have non-trivial instances and the



bounds are statements about nonzero quantities. What it does not prove: that the instantiated bounds are sharp, or that the witness geometry models any physical large-field configuration; it is a consistency certificate, not an application.

## 8 Related work and novelty

*Cluster expansions.* The abstract polymer framework and its convergence criteria go back to Kotecký–Preiss [1], with the modern sharp form of the criterion due to Fernández–Procacci [2]; pedagogical accounts include Brydges [3]. Tree-graph inequalities dominating Ursell coefficients by spanning-tree counts originate with Penrose [4]; the family of tree and forest formulas is surveyed by Abdesselam–Rivasseau [5]. The upstream repository formalizes a Kotecký–Preiss criterion and a Mayer–Ursell expansion, and the majorants  $S^\#$ ,  $T^\#$  of Definition 2.5 are the standard objects of this tradition.

*Holes and large fields.* The specific geometry — hole-respecting polymers, skeletons, and the modified (Steiner) metric — is that of renormalization-group analyses of lattice gauge theory in the Balaban school [6], in the concrete form of Dimock’s expository series [7, 8, 9]; the hole-component families of Definition 2.2 follow [8, §3.1.2]. We are not aware of published *machine-checked* bounds for expansions in this geometry prior to the present programme.

*The Catalan identity.* The count of labelled rooted plane trees underlying Theorem 3.1 is classical combinatorics at the level of [10], and we make no novelty claim for the enumeration itself. What we have not found in the literature is its statement in exactly this normalized child-factorial form over spanning trees of  $K_{n+1}$ , its use to replace the  $4^n$  shape budget inside an Ursell tree majorant with holes, or a formalization of either; readers aware of a prior source for the identity in this form are invited to point us to it.

*What is new here.* To our knowledge: (i) the formalized exact evaluation (Theorem 3.1) and its formalized corollary chain into hole-respecting leaf summation (Theorem 3.4); (ii) the machine-checked volume-free constant structure  $M^{2n+1}$  with the single-gate discipline (1); (iii) the machine-checked non-vacuity witness; and (iv) the certified-constants companion with committed pre-registered bands and a documented registration failure (Remark 5.1). The mathematical ingredients are classical; the contribution is the machine-checked integration.

*Formalization context.* The development builds on Mathlib [11]; the Catalan/plane-tree layer contributes upstream API (a `PlaneTree` companion prepared for separate submission to Mathlib).

## 9 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

07642720480157414db592fa85b626dafb71355b

(lakefile and manifest agree); `python-flint` 0.9.0 on CPython 3.12.6; `tectonic` 0.15.0.

Theorem-to-artifact map (statuses as in Section 3; all Lean files in module `MarkedRootedClosure` unless noted):



| Result          | Lean name                                      | File                                    | Status      |
|-----------------|------------------------------------------------|-----------------------------------------|-------------|
| Thm. 3.1        | <code>...TreeCatalanIdentity</code>            | <code>PaperTheorems.lean</code>         | exact       |
| Thm. 3.2        | <code>...TreeBound</code>                      | <code>PaperTheorems.lean</code>         | exact       |
| Thm. 3.3        | <code>markedRootLeafGeometricBound</code>      | <code>PaperTheorems.lean</code>         | exact       |
| Thm. 3.4        | <code>markedRootCatalanBound</code>            | <code>PaperTheorems.lean</code>         | exact       |
| Thm. 3.5        | <code>targetPreservingWeightedTreeBound</code> | <code>PaperTheorems.lean</code>         | exact       |
| §7 witnesses    | <code>nonvacuous_*</code>                      | <code>NonVacuity.lean</code>            | exact       |
| §5 table        | —                                              | <code>scripts/c1_admissibility_*</code> | certified   |
| §6 ratios       | —                                              | integer arithmetic in text              | paper-level |
| §6 consequences | —                                              | —                                       | paper-level |

From a clean clone of the repository at the release revision:

```
lake exe cache get          # Mathlib oleans
lake build MarkedRootedClosure
lake env lean MarkedRootedClosure/Oracle.lean # axiom oracle
python scripts/c1_admissibility_arb.py      # certified table
tectonic -X compile \
  papers/c1-rooted-tree-majorants/c1_rooted_tree_majorants.tex
```

Release facts (recorded from the run logs): `lake build MarkedRootedClosure` completed with exit code 0 in **8244 jobs** at the Lean-integration commit

4f95a5ad8154deef0ef38716073810072332fd37

(release tag `c1-v1.0` on the final manuscript commit). The axiom oracle (`lake env lean MarkedRootedClosure/Oracle.lean`; log committed as `scripts/c1_oracle_output.txt` at the same commit) reports, for each of the seven theorems — the five of Section 3 and the two non-vacuity instantiations of Section 7 — exactly

`[propext, Classical.choice, Quot.sound],`

the three standard classical axioms of Mathlib; no project axiom, no `sorry`.

Pin-drift audit (measured): relative to the module’s original `archive/UPSTREAM.lock` (upstream commit `4e45246a...`), the blob of `KP/RootedLeafSummation.lean` is byte-identical at the build revision, while `RG/AppendixFSecondUrsellLeafSummation.lean` drifted (+144/−15 lines: the `treeShape` refactor and the Catalan route; the geometric theorem’s statement is byte-unchanged). The Catalan chain (`KP/RootedCatalanExact.lean`, `KP/PlaneTreeCatalan.lean` and their dependencies) postdates the lock and is therefore not audited against it; for those files, and for everything else, the authority is the build itself: the theorems are compiled against the release revision, not the lock.

## 10 Conclusion

Five machine-checked theorems give the exact combinatorial value and the volume-free analytic envelope of rooted tree-graph majorants for polymer expansions with holes, with certified constants and a machine-checked consistency witness. The limitations are stated where they arise: the signed-coefficient-to-majorant passage is upstream and the activity-series algebra is paper-level; the Catalan gain is polynomial and moves no radius; the witness certifies satisfiability, not sharpness. Reasonable next uses are (a) consuming Theorem 3.4 inside the upstream programme wherever the  $4^n$  budget is currently paid, and (b) upstreaming the plane-tree API.

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